CHAPTER 9

ACCOUNTING RATIOS

CHAPTER ESSENCE AT A GLANCE:
Compute and interpret fundamental financial ratios across Liquidity, Solvency, Activity, and Profitability groups to diagnose corporate financial health and creditworthiness.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

- **Liquidity Testing Protocols:** Measures short-term debt paying ability: Current Ratio (Current Assets divided by Current Liabilities, benchmark 2:1) and Quick (Liquid) Ratio (Quick Assets divided by Current Liabilities, benchmark 1:1).
- **Solvency and Leverage Metrics:** Evaluates long-term financial stability: Debt-Equity Ratio (Long-term Debt divided by Shareholders' Funds), Proprietary Ratio, Total Assets to Debt Ratio, and Interest Coverage Ratio.
- **Activity and Asset Efficiency Ratios:** Measures velocity of resource utilization: Inventory Turnover Ratio, Trade Receivables Turnover Ratio, Trade Payables Turnover Ratio, and Working Capital Turnover Ratio.
- **Profitability and Return Rates:** Assesses capital productivity: Gross Profit Ratio, Operating Ratio, Operating Profit Ratio, Net Profit Ratio, and Return on Investment (ROI and ROCE).

ESSENTIAL ACCOUNTING PRINCIPLES & FINANCIAL FORMATS

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}, \quad \text{Quick Ratio} = \frac{\text{Current Assets} - (\text{Inventories} + \text{Prepaids})}{\text{Current Liabilities}}$$

Short-Term Enterprise Liquidity Benchmarks

$$\text{Debt-Equity Ratio} = \frac{\text{Long-Term Debts}}{\text{Shareholders' Funds}}, \quad \text{ICR} = \frac{\text{EBIT}}{\text{Interest on Long-Term Debt}}$$

Capital Structure Solvency and Coverage Metrics

$$\text{Inventory Turnover} = \frac{\text{Cost of Revenue from Operations}}{\text{Average Inventory}} \quad [\text{Expressed in Times}]$$

Merchandise Operational Turnover Velocity

$$\text{Return on Investment (ROI)} = \frac{\text{Net Profit Before Interest and Tax (EBIT)}}{\text{Capital Employed}} \times 100$$

Enterprise Overall Capital Productivity Rate

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

- **Alert:** Inventories and Prepaid Expenses are excluded when calculating Quick Assets from Current Assets.
- **Alert:** In Debt-Equity Ratio, 'Debt' includes only Long-Term Debts (non-current borrowings and provisions); Current Liabilities are strictly excluded.
- **Alert:** Operating Ratio + Operating Profit Ratio = 100 percent. If Operating Ratio is 80 percent, Operating Profit Ratio is strictly 20 percent.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. What are Liquidity Ratios? State the ideal benchmark standards for Current Ratio and Quick Ratio. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** Ratios measuring the firm's capacity to meet its short-term maturing financial obligations as they fall due. **[1 Mark]**

• **Benchmarks:** Current Ratio standard is 2:1; Quick (Liquid) Ratio standard is 1:1. **[1 Mark]**

2. Define Interest Coverage Ratio (ICR). What does a high ICR indicate to financial institutions? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** Net Profit Before Interest and Tax divided by Interest on Long-Term Debt (expressed in times). **[1 Mark]**

• **Significance:** Indicates the number of times operational earnings cover fixed interest charges; a higher ratio indicates stronger margin of safety for lenders. **[1 Mark]**

3. Current Assets of a firm are Rs 4,00,000 and Current Liabilities are Rs 2,00,000. State whether the Current Ratio will Increase, Decrease, or Not Change if: (a) Cash paid to Trade Creditors Rs 50,000, (b) Goods purchased for cash Rs 20,000, (c) Machinery purchased on credit of 3 months Rs 40,000. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **(a) Creditor Payment:** Current Assets decrease to Rs 3,50,000 and Current Liabilities decrease to Rs 1,50,000. New ratio = 2.33:1. Verdict: **Increases**. **[1 Mark]**

• **(b) Cash Purchase of Goods:** Stock increases by Rs 20,000 and Cash decreases by Rs 20,000 (net Current Assets unchanged, Liabilities unchanged). Verdict: **No Change**. **[1 Mark]**

• **(c) Machinery Bought on Short-term Credit:** Current Assets unchanged, Current Liabilities increase to Rs 2,40,000. New ratio = 1.67:1. Verdict: **Decreases**. **[1 Mark]**

4. From the following information, calculate Inventory Turnover Ratio: Total Sales Rs 5,00,000; Sales Returns Rs 20,000; Gross Profit 25 percent on Cost of Sales; Opening Inventory Rs 70,000; Closing Inventory Rs 90,000. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Net Sales:** Rs 5,00,000 - Rs 20,000 = Rs 4,80,000. **[0.5 Mark]**

• **Cost of Sales (COGS):** If Cost is 100, Sales = 125. COGS = Rs 4,80,000 * $\frac{100}{125}$ = Rs 3,84,000. **[1 Mark]**

• **Average Inventory:** $\frac{Opening Rs 70,000 + Closing Rs 90,000}{2}$ = Rs 80,000. **[0.5 Mark]**

• **Inventory Turnover:** Rs 3,84,000 divided by Rs 80,000 = 4.8 Times. **[1 Mark]**

5. From the following details, calculate Debt-Equity Ratio and Proprietary Ratio: Equity Share Capital Rs 6,00,000; Reserves and Surplus Rs 2,00,000; 10 percent Debentures Rs 4,00,000; Bank Overdraft Rs 1,00,000; Total Assets Rs 16,00,000. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Shareholders' Funds:** Capital Rs 6,00,000 + Reserves Rs 2,00,000 = Rs 8,00,000. Long-term Debt = Rs 4,00,000. **[1 Mark]**

• **Debt-Equity Ratio:** Long-Term Debt divided by Shareholders' Funds = Rs 4,00,000 divided by Rs 8,00,000 = 0.5:1. **[1 Mark]**

• **Proprietary Ratio:** Shareholders' Funds divided by Total Assets = Rs 8,00,000 divided by Rs 16,00,000 = 0.5:1 (or 50 percent). **[1 Mark]**

6. Distinguish between Operating Ratio and Operating Profit Ratio. State the mathematical relationship that connects both. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Operating Ratio:** (Operating Cost divided by Revenue from Operations) * 100. Measures the percentage of sales consumed by operating costs. **[1 Mark]**

• **Operating Profit Ratio:** (Operating Profit divided by Revenue from Operations) * 100. Measures operational profitability after meeting operating expenses. **[1 Mark]**

• **Mathematical Link:** Operating Ratio + Operating Profit Ratio = 100 percent. **[1 Mark]**

7. Net Profit after interest and tax of Surya Ltd is Rs 2,40,000. Rate of tax is 40 percent. The company has 10 percent Debentures of Rs 10,00,000. Shareholders' Funds are Rs 15,00,000. Calculate Return on Investment (ROI). [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Profit Before Tax (PBT):** Net Profit after tax Rs 2,40,000 represents 60 percent. PBT = Rs 2,40,000 divided by 0.60 = Rs 4,00,000. **[1 Mark]**

• **Profit Before Interest and Tax (EBIT):** Debenture Interest = 10 percent of Rs 10,00,000 = Rs 1,00,000. EBIT = Rs 4,00,000 + Rs 1,00,000 = Rs 5,00,000. **[1.5 Marks]**

• **Capital Employed:** Shareholders' Funds Rs 15,00,000 + Debentures Rs 10,00,000 = Rs 25,00,000. **[0.5 Mark]**

• **ROI Calculation:** (EBIT Rs 5,00,000 divided by Capital Employed Rs 25,00,000) * 100 = 20.0 percent. **[1 Mark]**